When are the coefficients of $B(B(B(x)))$ odd?

A proof of two conjectures in OEIS A396093

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Abstract

Substituting the power series $B(x) = x/(1-x)^2 = x + 2x^2 + 3x^3 + \cdots$ into itself twice gives the series $A(x) = B(B(B(x))) = x + 6x^2 + 33x^3 + 174x^4 + \cdots$, whose coefficients form the sequence A396093 of the On-Line Encyclopedia of Integer Sequences (OEIS). Paul D. Hanna conjectured two rules for when these coefficients are even. We prove both, and more: the coefficient of x^n is odd if and only if n leaves remainder 1, 3, 7 or 9 on division by 10. The proof works with coefficients modulo 2, where the triple substitution reduces to the fraction $(x + x^3 + x^7 + x^9)/(1 + x^{10})$. The result has also been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [2] is a large public database of integer sequences. Many entries also record *conjectures*: patterns that the contributor observed in the data but did not prove.

This paper is about entry A396093 [1], contributed by Paul D. Hanna in May 2026. It starts from the series

$$B(x) = \frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + 4x^4 + \cdots ,$$

whose coefficient of x^n is n. Substituting B into itself twice gives a new series,

$$A(x) = B(B(B(x))) = \sum_{n \geq 0} a(n)\, x^n = x + 6x^2 + 33x^3 + 174x^4 + 892x^5 + \cdots .$$

(Section 2 explains what substituting one series into another means.) The coefficients $a(n)$ form the sequence; its first terms, starting at $n = 0$, are 0, 1, 6, 33, 174, 892, 4480, 22149, 108144, 522685, 2505112, 11921919, … The entry reads:

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G.f.  A(x) = B(B(B(x))), the third iteration of B(x) = x/(1-x)^2.

Conjecture:  a(2*n) is even for n >= 1.

Conjecture:  a(2*n-1) is even iff n is of the form 5*k-2 (k >= 1) for n >=
1.
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— OEIS A396093, Paul D. Hanna, May 2026

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence.

1.2 The question

The two conjectures say when $a(n)$ is even. The first says that every term with an even index is even. The second says that a term with an odd index $2n-1$ is even exactly when n is one of $3, 8, 13, 18, \ldots$, that is, when the index is one of $5, 15, 25, 35, \ldots$. The table shows the parity of the first forty terms (1 means odd, 0 means even). The indices printed in bold are those that leave remainder 1, 3, 7 or 9 on division by 10.

n	0	**1**	2	**3**	4	5	6	**7**	8	**9**	10	**11**	12	**13**	14	15	16	**17**	18	**19**
$a(n)$ mod 2	0	**1**	0	**1**	0	0	0	**1**	0	**1**	0	**1**	0	**1**	0	0	0	**1**	0	**1**
n	20	**21**	22	**23**	24	25	26	**27**	28	**29**	30	**31**	32	**33**	34	35	36	**37**	38	**39**
$a(n)$ mod 2	0	**1**	0	**1**	0	0	0	**1**	0	**1**	0	**1**	0	**1**	0	0	0	**1**	0	**1**

The data suggest a single rule for all indices, which we prove.

1.3 What this paper proves

Main result (Theorem 3.1). *The coefficient $a(n)$ is odd if and only if n leaves remainder* 1, 3, 7 *or* 9 *on division by* 10.

Both of Hanna's conjectures follow at once (Corollary 3.2). The key step is a short calculation showing that, once every coefficient is replaced by its remainder modulo 2, the series $A(x)$ becomes the fraction

$$\frac{x + x^3 + x^7 + x^9}{1 + x^{10}} = (x + x^3 + x^7 + x^9)\,(1 + x^{10} + x^{20} + \cdots),$$

from which the pattern can be read off directly.

1.4 What was known

The OEIS entry also records a closed formula for $A(x)$ as a ratio of two polynomials (due to Vaclav Kotesovec), and a linear recurrence for $a(n)$ (due to Seiichi Manyama). For any sequence given by such a ratio, the remainders modulo 2 are known to repeat periodically from some point on: there are only finitely many possible "states" of the recurrence modulo 2. So a periodic parity pattern was to be expected, and in principle its period could be found by a finite computation from the recorded formula. We do not rely on that formula. The proof below works directly from the definition and shows *why* the period is 10. We are not aware of an earlier proof of the two conjectures.

1.5 Motivation

Substituting a series into itself is a basic operation: it describes what happens when a process is repeated, and iterates of simple series like B appear throughout combinatorics. Their coefficients quickly become large and irregular. Remainders modulo small numbers are often the first place where structure becomes visible, and explaining such patterns is a standard way of understanding a sequence. The example also shows a general technique: computing modulo 2 can collapse a complicated expression (here a ratio of two polynomials of degrees 7 and 8) into something transparent. Section 6 describes related questions.

1.6 How to read this paper

Section 2 explains the notions we need: power series, substitution, and remainders. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result and open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [4, 3]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1 + x + x^2 + \cdots$ is the *multiplicative inverse* of $1 - x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1 V(x) + u_2 V(x)^2 + u_3 V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod m$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

2.4 The Frobenius identity

Lemma 2.1 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. The first equality holds for polynomials by repeated use of $(a+b)^p = a^p + b^p$; it holds for power series because each coefficient of $U(x)^p$ involves only finitely many coefficients of U. □

For $p = 2$ this says that squaring a series modulo 2 doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod 2$. In particular, the square of any series has, modulo 2, no terms of odd degree.

3 Results

Theorem 3.1. *For every $n \geq 0$,*

$$a(n) \text{ is odd} \quad \Longleftrightarrow \quad n \equiv 1,\ 3,\ 7 \text{ or } 9 \pmod{10}.$$

Equivalently, modulo 2 the generating function is

$$A(x) \equiv \frac{x + x^3 + x^7 + x^9}{1 + x^{10}} = \sum_{j \geq 0} \left(x^{10j+1} + x^{10j+3} + x^{10j+7} + x^{10j+9}\right) \pmod{2}. \tag{1}$$

Corollary 3.2 (Hanna's conjectures)**.** *(a) $a(2n)$ is even for every $n \geq 0$.*

(b) For $n \geq 1$, $a(2n-1)$ is even if and only if $n = 5k - 2$ for some integer $k \geq 1$.

Proof. (a) An even number never leaves remainder 1, 3, 7 or 9 on division by 10. (b) An odd number $2n - 1$ leaves one of the remainders $1, 3, 5, 7, 9$. By Theorem 3.1, $a(2n-1)$ is even if and only if $2n - 1 \equiv 5 \pmod{10}$. This means $2n \equiv 6 \pmod{10}$, that is, $n \equiv 3 \pmod{5}$. For $n \geq 1$ these are the numbers $3, 8, 13, \ldots$, which are exactly the numbers $5k - 2$ with $k \geq 1$. □

4 Proofs

The idea in brief. We reduce all coefficients modulo 2. There, $B(x) = x/(1 + x^2)$, and substituting a fraction into B gives another fraction by a simple rule (Lemma 4.1). Applying the rule three times expresses $A = B \circ B \circ B$ as a fraction, and multiplying its numerator and denominator by $1 + x^2$ turns the denominator into $1 + x^{10}$.

From now on we work in $\mathbb{F}_2[[x]]$, that is, with coefficients modulo 2, and we keep the names A and B for the reduced series. Because reduction is compatible with substitution, the reduction of $A = B(B(B(x)))$ equals the triple substitution of the reduction of B. Since $-1 = 1$ in $\mathbb{F}_2$, and since $(1 + x)^2 = 1 + x^2$ there (Lemma 2.1),

$$B(x) = \frac{x}{(1-x)^2} = \frac{x}{(1+x)^2} = \frac{x}{1+x^2}.$$

Lemma 4.1. *Let P and D be series over $\mathbb{F}_2$ with $P(0) = 0$ and $D(0) = 1$. Then*

$$B\left(\frac{P}{D}\right) = \frac{P\,D}{D^2 + P^2}.$$

Proof. Put $T = P/D$. Then $B(T) = T/(1 + T^2)$. Multiplying the numerator and the denominator by D^2 gives $PD/(D^2 + P^2)$. The new denominator has constant term 1, so this fraction is again a power series. □

Proof of Theorem 3.1. Starting from $x = x/1$ and applying Lemma 4.1 three times (all arithmetic modulo 2):

series	numerator P	denominator D
$B(x)$	x	$1 + x^2$
$B(B(x))$	$x(1 + x^2) = x + x^3$	$(1 + x^2)^2 + x^2 = 1 + x^2 + x^4$
$B(B(B(x)))$	$(x + x^3)(1 + x^2 + x^4) = x + x^7$	$(1 + x^2 + x^4)^2 + (x + x^3)^2 = 1 + x^2 + x^4 + x^6 + x^8$

For instance, in the last row $(x+x^3)(1+x^2+x^4) = x+2x^3+2x^5+x^7$, and the terms with coefficient 2 vanish modulo 2. Hence

$$A(x) = \frac{x+x^7}{1+x^2+x^4+x^6+x^8}.$$

Now multiply the numerator and the denominator by $1+x^2$. The numerator becomes $(x+x^7)(1+x^2) = x+x^3+x^7+x^9$. The denominator becomes $(1+x^2)(1+x^2+x^4+x^6+x^8) = 1+x^{10}$, because every middle term appears twice and so vanishes. This proves the first form of (1). Finally, $1/(1+x^{10}) = 1+x^{10}+x^{20}+\cdots$, which gives the second form. Reading off coefficients, $a(n)$ is odd exactly when $n \equiv 1, 3, 7, 9 \pmod{10}$. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed $B(B(B(x)))$ directly from the definition. The result reproduces all 27 terms printed in the OEIS entry, and the parity rule of Theorem 3.1 holds for every $n < 600$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [7]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [5] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.19.0) on top of Mathlib [6], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A396093.lean` of the repository, in the namespace `OEIS.A396093`. There B is defined by the equation $B \cdot (1-x)^2 = x$ and A as $B \circ B \circ B$. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
$B(x) \cdot (1-x)^2 = x$ (definition of B)	`source_denominator`
$A = B \circ B \circ B$, and $a(n)$ is its coefficient of x^n	`source_composition`, `coefficient_definition`
Equation (1): $A \cdot (1+x^{10}) = x+x^3+x^7+x^9$ over $\mathbb{F}_2$	`reduced_denominator`
Theorem 3.1	`complete_parity`, `even_iff`
Corollary 3.2(a) and (b)	`even_indices`, `odd_indices`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. Theorem 3.1 describes the parity of every coefficient, so it settles both conjectures of the entry completely. It says nothing about the remainders of $a(n)$ modulo other numbers, or about the size of $a(n)$.

Other iterates. The same calculation applies to every iterate of B. The first two rows of the table above already show that $B(x) \equiv x/(1+x^2)$ and, after multiplying by $1+x^2$, that $B(B(x)) \equiv (x+x^5)/(1+x^6) \pmod 2$. So the coefficients of $B(B(x))$ (OEIS A030267) are odd exactly when $n \equiv 1$ or $5 \pmod 6$. For the fourth iterate, a computer calculation shows a parity pattern of period 34. We do not know a rule that gives the period for every iterate.

Higher moduli. In a computer calculation, the remainders of $a(n)$ modulo 4 repeat with period 20, starting from $n = 0$; we have not proved that this continues. Since A is a ratio of polynomials, these remainders are periodic from some point on, and the method of this paper could be used to determine the period.

About this paper

How the work was done. This paper is one of 29 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [7]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [7], where the Lean sources and the numerical checks are also available.

Acknowledgements

The sequence and both conjectures are due to Paul D. Hanna [1]. The closed formula and recurrence mentioned in the introduction were contributed to the entry by Vaclav Kotesovec and Seiichi Manyama.

References

[1] P. D. Hanna, entry A396093 (Third iteration of $x/(1-x)^2$), *The On-Line Encyclopedia of Integer Sequences*, May 2026. `https://oeis.org/A396093`. Text as of October 2026.

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[7] D. D. Goyal, *Proofs and corrections of OEIS conjectures*, repository, 2026. `https://github.com/dvgyl/oeis-a396843-proof`.